

Programming Tech Domains Guide

1. Web Development

- **HTML & CSS (2-4 weeks)**
 - Resources: [MDN Web Docs](#), [FreeCodeCamp](#)
- **JavaScript (4-6 weeks)**
 - Resources: [JavaScript.info](#), [Eloquent JavaScript \(book\)](#)
- **Front-End Frameworks:**
 - **React (4-6 weeks)**
 - Resources: [React Official Documentation](#)
 - **Angular (4-6 weeks)**
 - Resources: [Angular Official Documentation](#)
 - **Vue.js (3-5 weeks)**
 - Resources: [Vue.js Official Documentation](#)
- **Back-End Development:**
 - **Node.js (4-6 weeks)**
 - Resources: [Node.js Documentation](#)
 - **Django (4-6 weeks)**
 - Resources: [Django Official Documentation](#)
 - **Ruby on Rails (4-6 weeks)**
 - Resources: [Ruby on Rails Guides](#)
- **Databases:**
 - **SQL (MySQL/PostgreSQL) (3-5 weeks)**
 - Resources: [MySQL Documentation](#), [PostgreSQL Documentation](#)
 - **NoSQL (MongoDB) (3-5 weeks)**
 - Resources: [MongoDB Documentation](#)
- **Advanced Topics:**
 - **Web Performance (2-4 weeks)**
 - Resources: [Google Web Fundamentals](#)
 - **Security (2-4 weeks)**
 - Resources: [OWASP](#)
 - **Responsive Design (2-4 weeks)**
 - Resources: [FreeCodeCamp Responsive Web Design](#)

2. Mobile Development

- **Basic Programming:**
 - **Swift (4-6 weeks)**
 - Resources: [Apple Developer Documentation](#), [Swift Playgrounds \(app\)](#)
 - **Kotlin (4-6 weeks)**
 - Resources: [Kotlin Documentation](#)
- **Development Environment:**
 - **Xcode (2-4 weeks)**
 - Resources: [Apple Developer Xcode Documentation](#)
 - **Android Studio (2-4 weeks)**
 - Resources: [Android Studio Documentation](#)
- **UI/UX Design (3-5 weeks)**
 - Resources: [Material Design Guidelines](#), [Human Interface Guidelines](#)
- **APIs & Data Handling (3-5 weeks)**
 - Resources: [Google APIs Documentation](#), [Apple Developer API Documentation](#)
- **Cross-Platform Development:**
 - **Flutter (4-6 weeks)**
 - Resources: [Flutter Official Documentation](#)
 - **React Native (4-6 weeks)**
 - Resources: [React Native Official Documentation](#)

3. Data Science

- **Python/R Basics:**
 - **Python (3-5 weeks)**
 - Resources: [Python Official Documentation](#), [Kaggle Learn Python](#)
 - **R (3-5 weeks)**
 - Resources: [R Documentation](#), [R for Data Science \(book\)](#)
- **Data Manipulation:**
 - **Pandas (3-5 weeks)**
 - Resources: [Pandas Documentation](#)
 - **dplyr (R) (3-5 weeks)**
 - Resources: [dplyr Documentation](#)
- **Statistical Analysis (3-5 weeks)**
 - Resources: [Khan Academy Statistics](#)
- **Machine Learning (4-6 weeks)**
 - Resources: [Scikit-Learn Documentation](#), [TensorFlow Documentation](#)
- **Data Visualization:**
 - **Matplotlib/Seaborn (Python) (3-5 weeks)**
 - Resources: [Matplotlib Documentation](#), [Seaborn Documentation](#)
 - **ggplot2 (R) (3-5 weeks)**
 - Resources: [ggplot2 Documentation](#)

4. Machine Learning/AI

- **Foundations (3-5 weeks)**
 - Resources: [Coursera Machine Learning by Andrew Ng](#)
- **Python Libraries:**
 - **TensorFlow (4-6 weeks)**
 - Resources: [TensorFlow Documentation](#)
 - **PyTorch (4-6 weeks)**
 - Resources: [PyTorch Official Documentation](#)
 - **Keras (3-5 weeks)**
 - Resources: [Keras Documentation](#)
- **Algorithms (4-6 weeks)**
 - Resources: [Pattern Recognition and Machine Learning \(book\)](#)
- **Projects (4-6 weeks)**
 - Resources: [Kaggle Competitions](#)
- **Specializations:**
 - **NLP (4-6 weeks)**
 - Resources: [Hugging Face Transformers Documentation](#)
 - **Computer Vision (4-6 weeks)**
 - Resources: [OpenCV Documentation](#)

5. Cloud Computing

- **Basic Concepts (2-4 weeks)**
 - Resources: [Cloud Computing Basics by IBM](#)
- **Cloud Providers:**
 - **AWS (4-6 weeks)**
 - Resources: [AWS Training and Certification](#)
 - **Azure (4-6 weeks)**
 - Resources: [Microsoft Learn for Azure](#)
 - **Google Cloud (4-6 weeks)**
 - Resources: [Google Cloud Training](#)
- **Core Services:**
 - **Compute/Storage/Networking (4-6 weeks)**
 - Resources: [AWS Documentation](#), [Azure Documentation](#), [Google Cloud Documentation](#)
- **Deployment (3-5 weeks)**
 - Resources: [AWS Elastic Beanstalk Documentation](#), [Azure App Services Documentation](#)
- **Advanced Topics:**
 - **Serverless Computing (3-5 weeks)**
 - Resources: [AWS Lambda Documentation](#), [Azure Functions Documentation](#)
 - **Kubernetes (4-6 weeks)**
 - Resources: [Kubernetes Documentation](#)

Programming Tech Domains Guide (Last 5)

6. Cybersecurity

- **Foundations (2-4 weeks)**
 - Resources: [Cybrary](#), [Khan Academy Cybersecurity](#)
- **Network Security (3-5 weeks)**
 - Resources: [Cisco Networking Academy](#), [CompTIA Network+](#)
- **Ethical Hacking (4-6 weeks)**
 - Resources: [TryHackMe](#), [Hack The Box](#)
- **Cryptography (3-5 weeks)**
 - Resources: [Coursera Cryptography I](#), [Khan Academy Cryptography](#)
- **Web Application Security (3-5 weeks)**
 - Resources: [OWASP](#), [PortSwigger Web Security Academy](#)
- **Security Tools (4-6 weeks)**
 - Resources: [Kali Linux](#), [Wireshark](#)

7. Blockchain Development

- **Blockchain Basics (2-4 weeks)**
 - Resources: [Blockchain by IBM](#), [Coursera Blockchain Basics](#)
- **Ethereum Development (4-6 weeks)**
 - Resources: [Ethereum Documentation](#), [Solidity Documentation](#)
- **Smart Contracts (3-5 weeks)**
 - Resources: [CryptoZombies](#), [Dapp University](#)
- **Decentralized Applications (DApps) (3-5 weeks)**
 - Resources: [Truffle Suite](#), [Web3.js Documentation](#)
- **Blockchain Security (3-5 weeks)**
 - Resources: [Consensys Blockchain Security](#), [Smart Contract Security](#)

8. Game Development

- **Game Development Basics (2-4 weeks)**
 - Resources: [GameDev.tv](#), [Coursera Game Design](#)
- **Unity (4-6 weeks)**
 - Resources: [Unity Learn](#), [Unity Documentation](#)

- **Unreal Engine (4-6 weeks)**
 - Resources: [Unreal Online Learning](#), [Unreal Engine Documentation](#)
- **2D/3D Graphics (3-5 weeks)**
 - Resources: [Blender Guru](#), [Godot Engine](#)
- **Game Physics (3-5 weeks)**
 - Resources: [Physics for Game Developers \(book\)](#), [GDC Vault](#)
- **Multiplayer Development (3-5 weeks)**
 - Resources: [Photon Engine](#), [Steamworks](#)

9. DevOps

- **DevOps Foundations (2-4 weeks)**
 - Resources: [Coursera DevOps](#), [AWS DevOps](#)
- **Version Control (2-4 weeks)**
 - Resources: [Git Official Documentation](#), [Pro Git \(book\)](#)
- **CI/CD (4-6 weeks)**
 - Resources: [Jenkins Documentation](#), [CircleCI](#)
- **Containerization (4-6 weeks)**
 - Resources: [Docker Documentation](#), [Kubernetes Documentation](#)
- **Infrastructure as Code (3-5 weeks)**
 - Resources: [Terraform Documentation](#), [Ansible Documentation](#)
- **Monitoring & Logging (3-5 weeks)**
 - Resources: [Prometheus](#), [ELK Stack](#)

10. Software Development with APIs

- **API Basics (2-4 weeks)**
 - Resources: [RESTful API Tutorial](#), [Postman Learning Center](#)
- **API Design & Development (3-5 weeks)**
 - Resources: [Swagger](#), [API Gateway Documentation](#)
- **Authentication & Security (3-5 weeks)**
 - Resources: [OAuth 2.0 Documentation](#), [JWT \(JSON Web Tokens\)](#)
- **GraphQL (3-5 weeks)**
 - Resources: [GraphQL Documentation](#), [Apollo GraphQL](#)
- **API Testing (2-4 weeks)**
 - Resources: [Postman](#), [SoapUI](#)
- **Advanced API Topics (4-6 weeks)**
 - Resources: [Rate Limiting](#), [API Versioning](#)